



香港中文大學(深圳)

The Chinese University of Hong Kong, Shenzhen

**Introduction to Data Science**

# **Lecture 10: Regression Analysis**

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# Recap



## Probability

- When the true **model** is **known**, you can **predict** the outcomes of the **samples**.

## Statistics

- By **analyzing** the generated **samples**, you can **estimate** the underlying true **model**.

# Statistics

- Target: **extract some useful information** from the data.
- What you need?
  - Samples
  - A set of possible models
  - A criterion for quantifying model performance

# Maximum Likelihood Estimation (MLE)

- Target: estimate  $\theta$  of a model
- Samples:  $X_1, X_2 \dots, X_n$
- Possible models:  $\theta \in \Theta$  (additional information)
- Model performance (probability):  $L(\theta) = P(X_1, X_2 \dots, X_n | \theta)$
- Estimator:  $\hat{\theta}$  such that  $L(\theta)$  is maximized at  $\theta = \hat{\theta}$ .

# Formal Definition

- Given a model with an unknown parameter  $\theta$
- Given samples:  $X_1, X_2 \dots, X_n$
- The probability that a model ( $\theta$ ) generates the samples is called **likelihood**.  $L(\theta) = P(X_1, X_2 \dots, X_n | \theta)$
- To construct the right statistic, we choose  $\hat{\theta}$  such that  $L(\theta)$  is maximized at  $\theta = \hat{\theta}$ . **Maximum Likelihood Estimation (MLE)**

# Likelihood function

- Given a model with an unknown parameter  $\theta$
- Given samples:  $X_1, X_2 \dots, X_n$

## Continuous RV model:

- Likelihood:  $L(\theta) = \prod_i f(X_i | \theta)$
- Log-Likelihood:  $l(\theta) = \sum_i \log(f(X_i | \theta))$

$f$ : the model's probability density function (PDF)

## Discrete RV model:

- Likelihood:  $L(\theta) = \prod_i P(X_i | \theta)$
- Log-Likelihood:  $l(\theta) = \sum_i \log(P(X_i | \theta))$

$P$ : the model's probability mass function (PMF)

# Likelihood function

- Given a model with an unknown parameter  $\theta$
- Given samples:  $X_1, X_2, \dots, X_n$

## Continuous RV model:

- Likelihood:  $L(\theta) = \prod_i f(X_i | \theta)$
- Log-Likelihood:  $l(\theta) = \sum_i \log(f(X_i | \theta))$

Why do we multiply the pdfs or pmfs here?

Because we assume those samples are **independent** observations

## Discrete RV model:

- Likelihood:  $L(\theta) = \prod_i P(X_i | \theta)$
- Log-Likelihood:  $l(\theta) = \sum_i \log(P(X_i | \theta))$



# Example: no additional information

- For drug experiments,
  - **N** experiments
  - **M** successes
- Possible models: Bernoulli distribution with probability  $p$ .
- Given one model, the probability of generating such samples:  
 $L(p) = p^M (1 - p)^{N-M}$
- Most likely: the model **maximizes** the probability  $L(p)$

# Example: no additional information

- Given one model, the probability of generating such samples:

$$L(p) = p^M (1 - p)^{N-M}$$

- Maximize the probability



We often maximize the **logarithm** of the probability (usually easier to maximize) of generating such samples

$$l(p) = \log L(p) = M * \log p + (N-M) * \log (1-p)$$

# Example: no additional information

- To maximize a continuous function  $l(p)$ 
  - Find the point  $l'(p) = 0$  and  $l''(p) < 0$
  - $p \in [0,1]$

$$l(p) = \log L(p) = M * \log p + (N-M) * \log (1-p)$$

$$\bullet \quad l'(p) = \frac{M}{p} - \frac{N-M}{1-p} = 0 \quad \longrightarrow \quad p = M/N \text{ (sample cure rate)}$$

# Example: with additional information

- You may have additional information.
  - Drug 1:  $p = 0.8$
  - Drug 2:  $p = 0.9$
  - Samples:  $M=81$  successes from  $N = 100$  experiments.
- Potential models: Bernoulli with  $p \in \{0.8, 0.9\}$

$$l = \log L = M \cdot \log p + (N-M) \cdot \log (1-p)$$

- Drug 1: -48.6539
- Drug 2: -52.2833

**Conclusion: drug 1 is more likely to be experimented.**

# An Additional Example of MLE

## Baseball Team

- The weights of players in a baseball team are  
 $\{150, 143, 132, 160, 175, 190, 123, 154\}$
- Assume their weights are **uniformly** distributed over an interval  $[a, b]$
- What are good estimators for  $a$ ? for  $b$ ?

This example will show that the **MLE could be complicated to solve**, e.g., the equation  $l'(\theta) = 0$  may be difficult to solve, or it may **not** always be possible to use calculus methods **directly** to find the maximum of  $L(\theta)$ .

# MLE: Uniform

Let  $X$  be a Uniform random variable on the interval  $[0, \theta]$

$$f(x; \theta) = \begin{cases} \frac{1}{\theta}, & \text{for } 0 \leq x \leq \theta, \\ 0, & \text{otherwise,} \end{cases} = \frac{1}{\theta} \mathbf{1}_{\{0 \leq x \leq \theta\}}$$

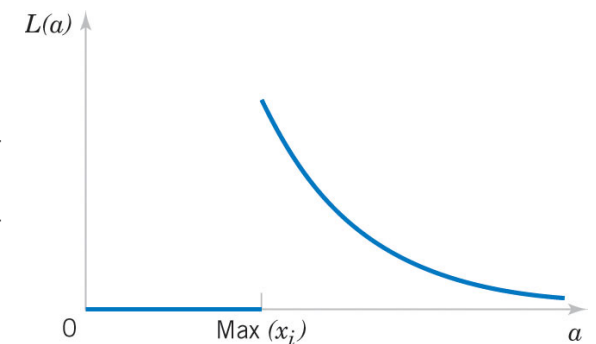
indicator function  $\mathbf{1}_A(x)$

$$\mathbf{1}_A(x) = \begin{cases} 1, & \text{if } x \in A, \\ 0, & \text{otherwise} \end{cases}$$

The likelihood function of a random sample of size  $n$  is:

$$L(\theta) = \prod_{i=1}^n f(x_i; \theta) = \frac{1}{\theta^n} \prod_{i=1}^n \mathbf{1}_{\{0 \leq x_i \leq \theta\}} = \begin{cases} \frac{1}{\theta^n}, & \text{if } \theta \geq \max\{X_1, X_2, \dots, X_n\} \\ 0, & \text{if } \theta < \max\{X_1, X_2, \dots, X_n\} \end{cases}$$

$$\hat{\theta} = \max\{X_1, X_2, \dots, X_n\}$$



Calculus methods don't work here because  $L(\theta)$  is maximized at the **discontinuity**. Clearly,  $\theta$  cannot be smaller than  $\max(x_i)$ , thus the MLE is  $\max\{X_1, X_2, \dots, X_n\}$ .

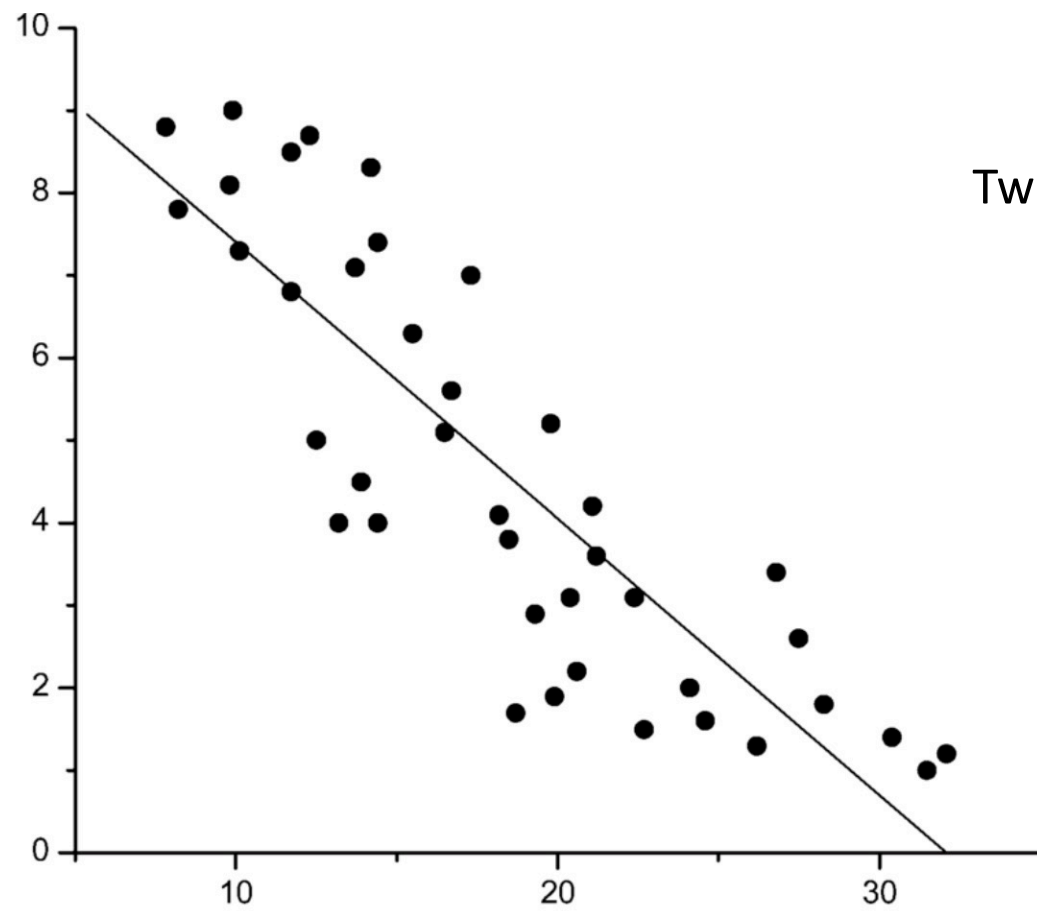
# More discussion on MLE

- Is MLE of the mean of a distribution always the sample average?
- Not always!
- True for Normal distribution, Bernoulli distribution...
- False for uniform distribution...

# Linear regression

Find the relationship between  $X$  and  $Y$



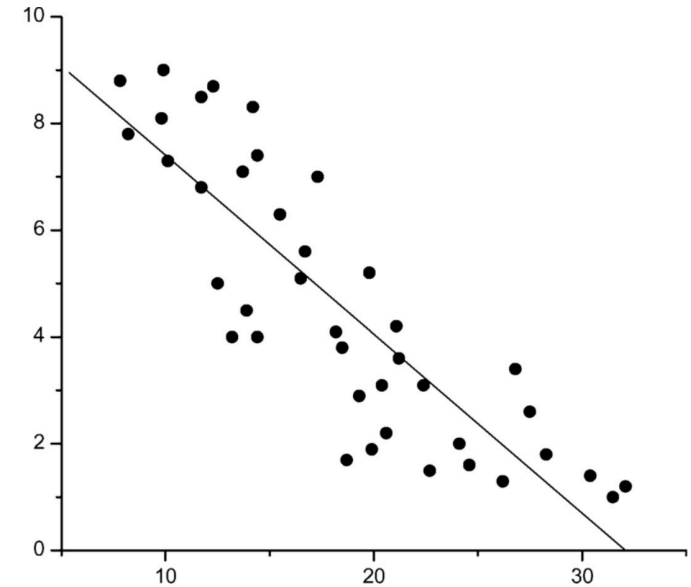


Two Stocks (X,Y)

# Correlation

- With a positive correlation (positively correlated),
  - Larger  $x$  implies larger  $y$  (vice versa)
- With zero correlation (uncorrelated),
  - No clear relationship between  $x$  and  $y$
- With a negative correlation (negatively correlated),
  - Larger  $x$  implies smaller  $y$  (vice versa)

- Negative: larger  $x$  implies smaller  $y$ .

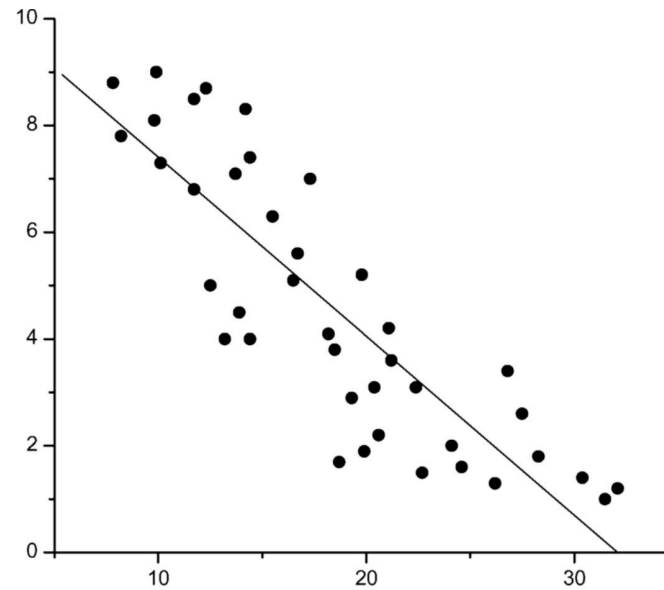


- **Question:** when  $x$  increases by a certain quantity, what's the reduction in  $y$ ?
- Use a line to approximate the relationship:
  - Regression analysis.

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$$Y \approx \beta_0 + \beta_1 X$$

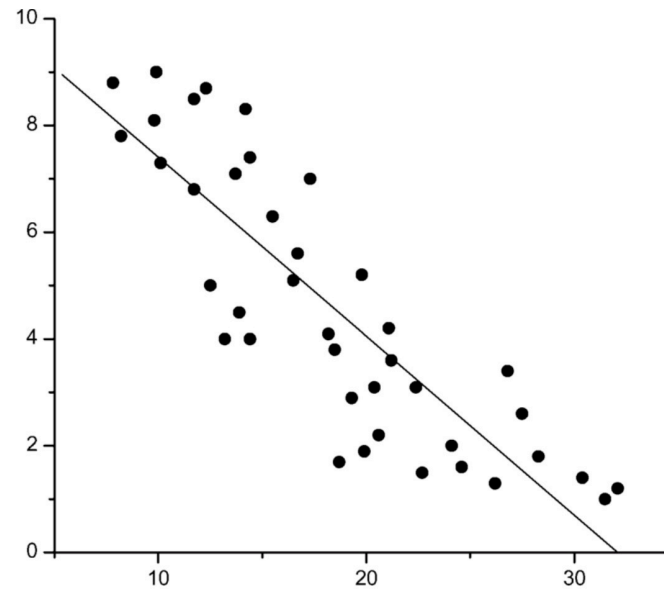
$$Y \sim N(\beta_0 + \beta_1 X, \sigma^2)$$



- Use a line to approximate the relationship:
  - Regression analysis: choose the best  $\beta_0, \beta_1, \sigma^2$  using MLE

$$Y \approx \beta_0 + \beta_1 X$$

$$Y \sim N(\beta_0 + \beta_1 X, \sigma^2)$$



- PDF for normal

$$\frac{1}{\sqrt{2\pi}\sigma} \exp \left[ -\frac{1}{2} \left( \frac{x - \mu}{\sigma} \right)^2 \right].$$

- Samples:  $(X_1, Y_1), \dots, (X_N, Y_N)$

- For the model with  $\beta_0, \beta_1, \sigma^2$ , the likelihood is

$$\frac{1}{(\sqrt{2\pi})^n \sigma^n} \exp \left[ -\frac{1}{2} \frac{\sum_i (Y_i - \beta_1 X_i - \beta_0)^2}{\sigma^2} \right]$$

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- Given  $\sigma^2$ , to **maximize** the likelihood, we only need to **minimize**

$$\sum_i (Y_i - \beta_1 X_i - \beta_0)^2$$

- Taking derivative over  $\beta_0$  and  $\beta_1$ , we have

$$\sum_i (Y_i - \beta_1 X_i - \beta_0) = 0$$

$$\sum_i (Y_i - \beta_1 X_i - \beta_0) X_i = 0$$

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- Given  $\sigma^2$ , to **maximize** the likelihood, we only need to **minimize**

$$\sum_i (Y_i - \beta_1 X_i - \beta_0)^2$$

From High School:  
Least square regression  
最小二乘法

- Taking derivative over  $\beta_0$  and  $\beta_1$ , we have

$$\sum_i (Y_i - \beta_1 X_i - \beta_0) = 0$$

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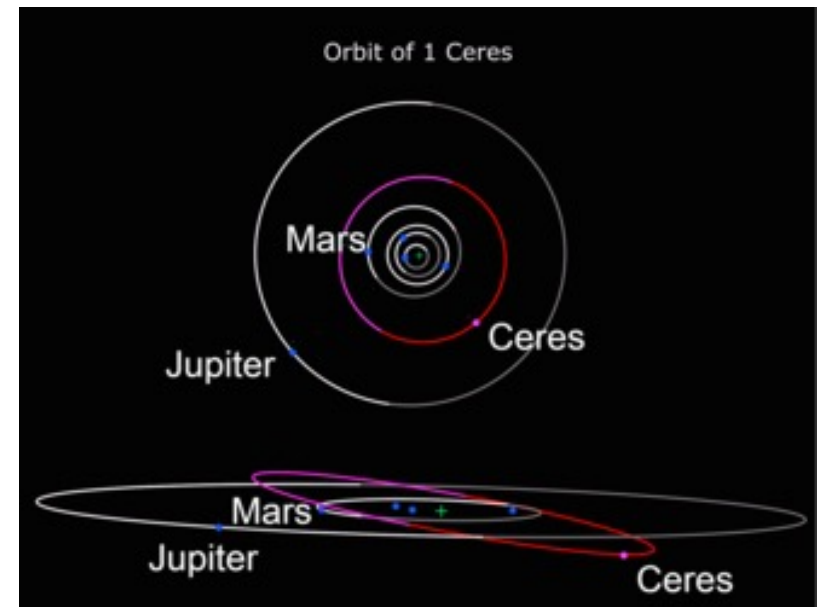


# Interesting history (Optional)

## The Problem: Predicting the Orbit of the Asteroid Ceres

In 1801, the astronomer **Giuseppe Piazzi** discovered an asteroid between Mars and Jupiter, which he named **Ceres**. At the time, astronomers were unable to accurately predict its orbit due to the small size of the asteroid and the lack of precise observational data. Piazzi's initial observations were limited and did not fully determine Ceres's orbit, making it difficult to calculate where it would be in the sky after a few months.

In 1809, **Gauss** took on the problem of predicting the orbit of Ceres. He was tasked with using only the sparse observational data available (fewer than 10 measurements) to determine the asteroid's orbit and predict its future position in the sky. Gauss's method was groundbreaking because, instead of simply using the available measurements directly, he took into account **errors in the data** and developed a new technique for fitting the orbit to the noisy observations. This method is now known as **the method of least squares**.



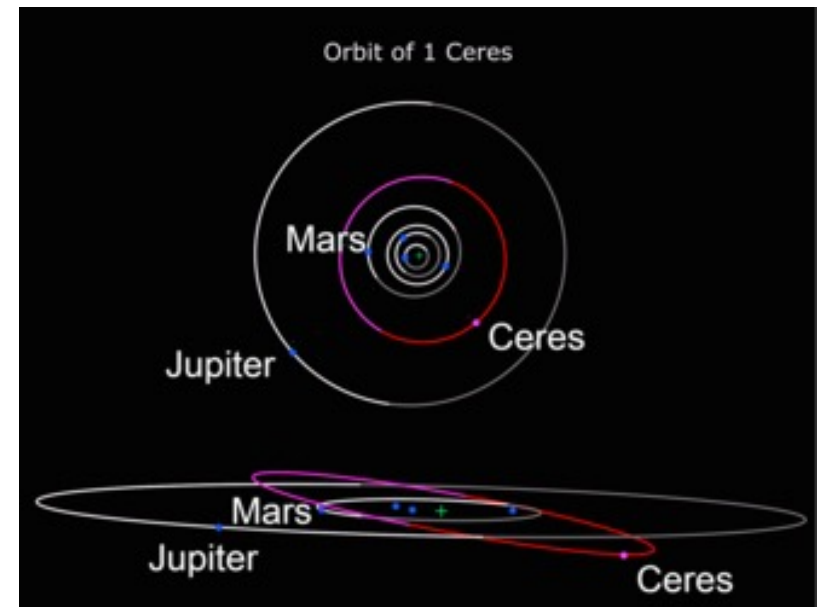
# Interesting history (Optional)

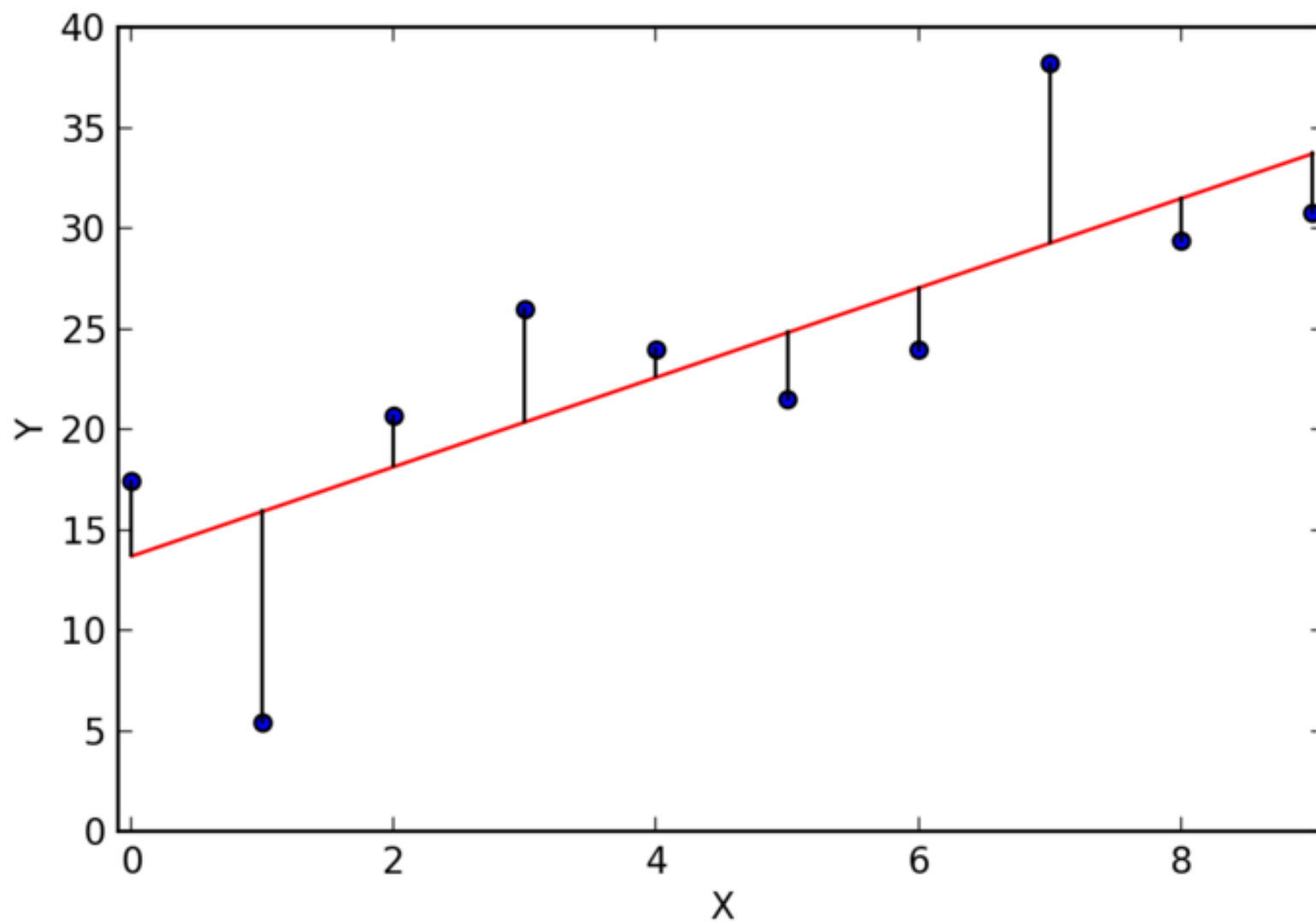
## Gauss's Method of Least Squares

Gauss's method was a breakthrough because it accounted for the fact that astronomical measurements are never perfect. Observations are inherently **noisy** due to instrumental errors, observational errors, and other random factors. Gauss realized that instead of discarding "outlier" data points, it would be more effective to **fit a model** (in this case, the orbit of Ceres) to all the available data, while minimizing the overall errors.

This technique is called the **least squares method**, and it works by finding the best-fitting curve (or line) that minimizes the sum of the **squared deviations** between the observed data and the predicted values. Gauss used this approach to successfully predict Ceres's position, which was later confirmed when astronomers observed Ceres exactly where Gauss had predicted.

However, Gauss's work on orbital prediction didn't just end with the development of the least squares method. He also recognized that measurement errors were not randomly distributed—they followed a pattern.





$$\Sigma_i(Y_i - \beta_1 X_i - \beta_0)X_i = 0 \text{ AND } \Sigma_i(Y_i - \beta_1 X_i - \beta_0) = 0$$

Eliminate  $\beta_0$  first:

$$\Sigma_i(Y_i - \beta_1 X_i - \beta_0) = 0 \rightarrow \beta_0 = \frac{1}{N} \Sigma_i(Y_i - \beta_1 X_i) = \bar{Y} - \beta_1 \bar{X}$$

$$\text{MLE: } \left\{ \begin{array}{l} \widehat{\beta}_1 = \frac{\Sigma_i(X_i - \bar{X})(Y_i - \bar{Y})}{\Sigma_i(X_i - \bar{X})^2} \\ \widehat{\beta}_0 = \bar{Y} - \widehat{\beta}_1 \bar{X} \end{array} \right.$$

# Is our regression model correct?

- The model we propose

$$Y \sim N(\beta_0 + \beta_1 X, \sigma^2) \text{ or } Y - \beta_0 - \beta_1 X \sim N(0, \sigma^2)$$

Linear regression assumes that...

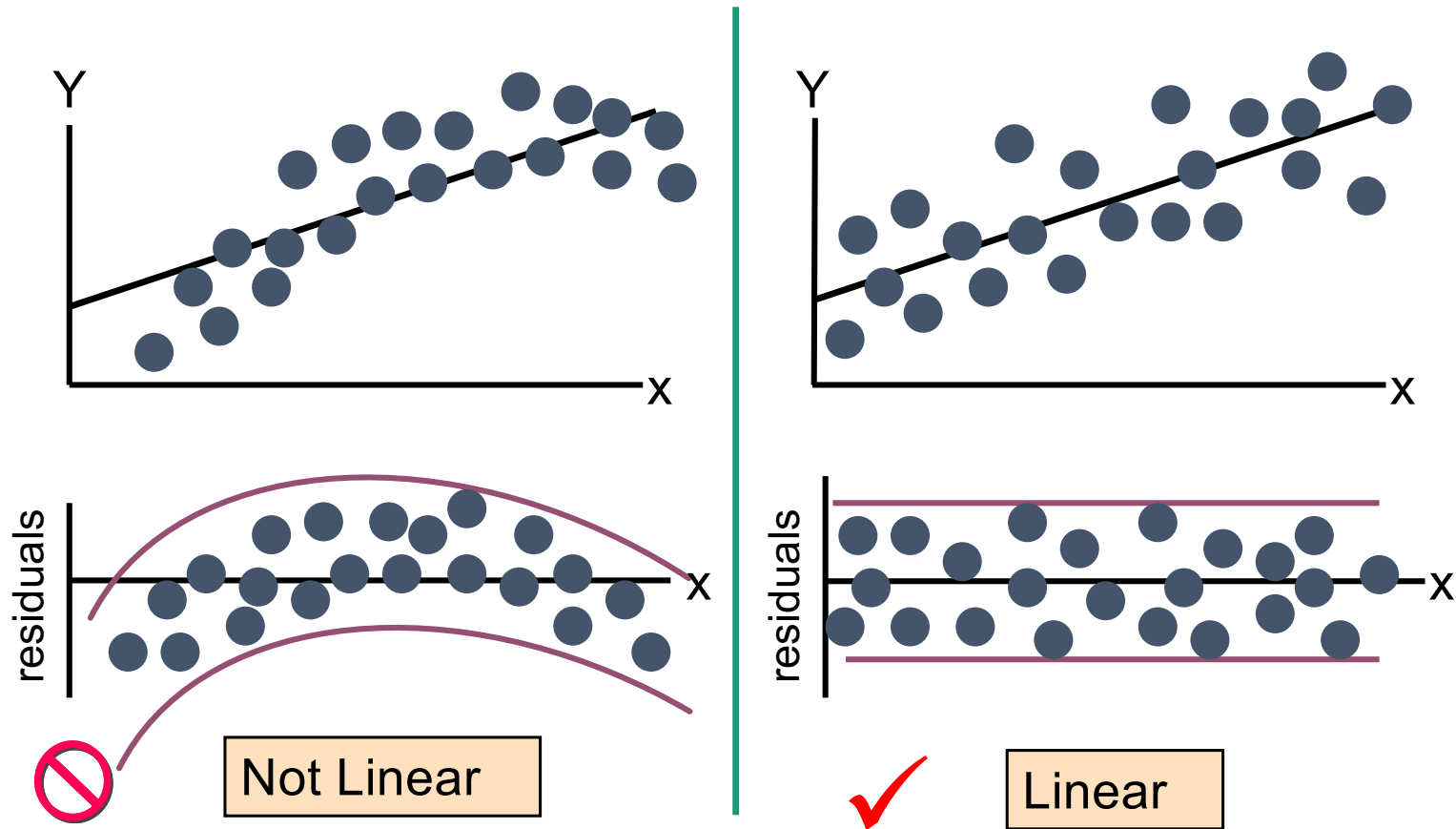
1. The relationship between X and Y is **linear**
2. The variance of  $Y - \beta_0 - \beta_1 X$  at every value of X is the **same** (homogeneity of variances)

# Residual Analysis: check assumptions

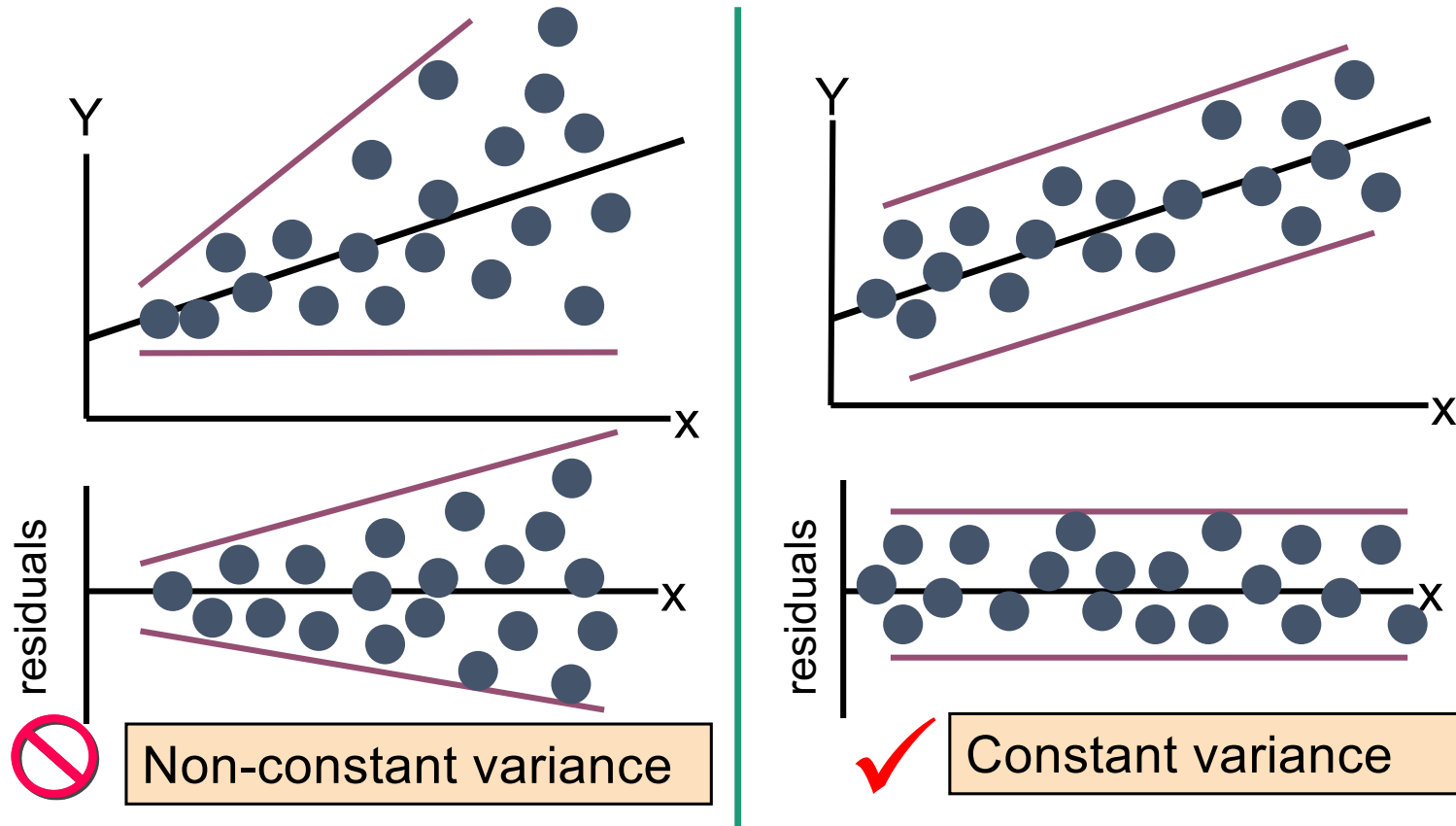
$$\text{Residual: } e_i := Y_i - \widehat{\beta}_0 - \widehat{\beta}_1 X_i$$

- Check the assumptions by examining the residuals
  - Examine for linearity assumption:
    - *$e_i$  does not depend on  $X_i$*
  - Evaluate constant-variance assumption:
    - *variance of  $e_i$  does not depend on  $X_i$*
- Graphical Analysis of Residuals: Can plot residuals vs.  $X$

# Residual Analysis for Linearity



# Residual Analysis for constant-variance





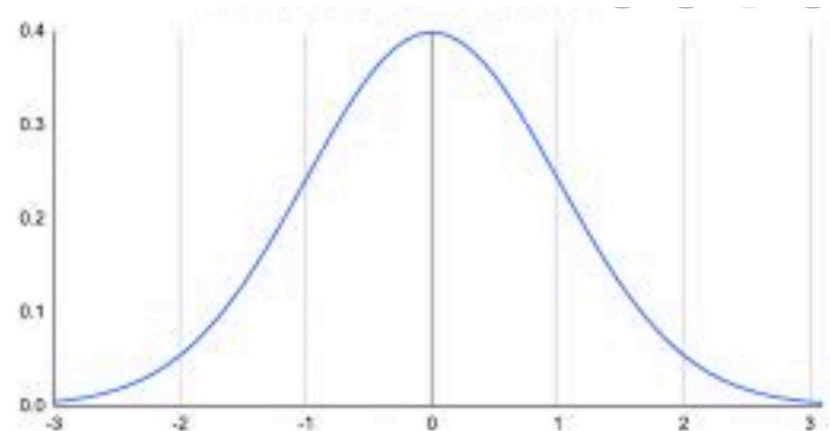
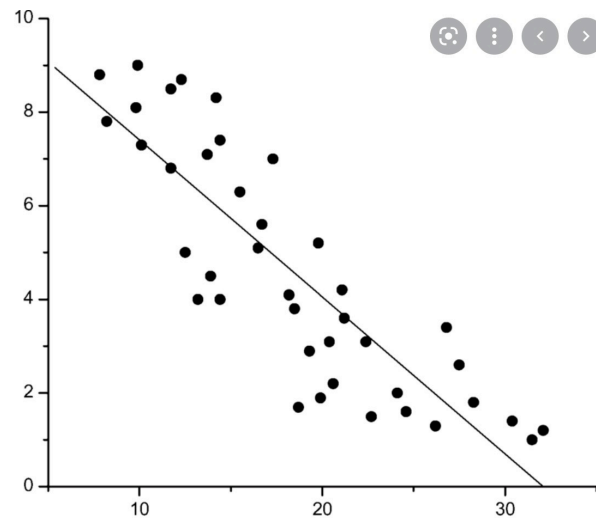
# Summary of Regression Analysis:

- Target: find a relationship between  $X$  and  $Y$ .
  - Proposed models:  $Y \sim N(\beta_0 + \beta_1 X, \sigma^2)$ .
  - Choose the best parameters: MLE.
  - Check whether the model is acceptable: Residual analysis.

# 1. Why we propose $Y \sim N(\beta_0 + \beta_1 X, \sigma^2)$ ?

From data, we observe that

- They are more likely to be linearly dependent with each other.
- $Y$  is centralized at some value  $\beta_0 + \beta_1 X$ .



## 2. Why we use MLE to estimate parameters?

- Among the proposed models, choose the one with the **largest probability** that the samples are generated.
- Recall example
  - Drug 1:  $p = 0.8$
  - Drug 2:  $p = 0.9$
  - Samples:  $M = 81$  successes from  $N=100$  experiments
  - Which drug do you think is experimented?

### 3. Whether the model is acceptable?

- The model assumption may be incorrect.
  - The relationship between  $X$  and  $Y$  is linear
  - The variance of  $Y - \beta_0 - \beta_1 X$  at every value of  $X$  is the same (homogeneity of variances)
- Residual analysis

